Goal 1: How Long the Data Generation Takes

DeepSeek-R1-Distill-Qwen-7B · 8 GPUs (RTX A6000, 48 GB each)

WHAT WE ARE GENERATING

We want to know whether a model has already decided its answer part-way through its reasoning. So we let it reason, cut that reasoning off at five points (10%, 30%, 50%, 70%, 90%), and restart it 100 times from each point to see where it ends up. If the 100 restarts agree, it had committed; if they scatter, it had not.

Those restarts are the cost. We do this for 3 reasoning attempts per question.

TIME REQUIRED

dataset	questions	restarts	GPU-hours	real time	status
MATH-500	500	691,900	~28	~5 hours	running
AIME-2025	30	36,500	~7	~1 hour	next
GPQA-Diamond	198	264,000	~43	~5 hours	deferred

8 GPUs run at once, so ~28 GPU-hours of work finishes in about 5 hours of real time.

Currently running (MATH-500 + AIME): about 6 hours.

Adding GPQA-Diamond would make it about 10 hours.

WHY IT IS MUCH FASTER THAN WE FIRST THOUGHT

Our first estimate, scaled up from small trial runs, was about 48 hours. Before committing the machines we ran a short test, which found three fixable problems — in one of them the GPUs were reserving memory but doing no work at all. Fixing them made the job roughly 5x cheaper.

MATH-500 figures are measured on the run in progress; the others are projected.

How the estimate was calculated

Time a small trial run, scale it up, then re-time the real thing

STEP 1 — TIME A SMALL TRIAL

We ran 33 questions per dataset with 10 restarts each on a single GPU and timed it.

dataset	restarts	time taken	restarts per second
MATH-500	4,470	53m 25s	1.39
GPQA-Diamond	4,400	3h 30m	0.35
AIME-2025	2,050	1h 55m	0.30

GPQA and AIME are ~4x slower simply because they write much more text per restart.

STEP 2 — SCALE UP TO ALL QUESTIONS AND 100 RESTARTS

Not every reasoning attempt is usable: some run past the length limit before they finish, and those cannot be cut up. The usable ones give us the restart counts.

dataset	questions	attempts	usable	restarts to write
MATH-500	500	1,500	1,442	691,900
AIME-2025	30	90	73	36,500
GPQA-Diamond	198	~594	~528	264,000

Restarts divided by the trial speed gave our first estimate:

MATH-500	691,900 / 1.39 per sec	=	138 GPU-hours
GPQA-Diamond	264,000 / 0.35 per sec	=	210 GPU-hours
AIME-2025	36,500 / 0.30 per sec	=	34 GPU-hours
TOTAL			382 GPU-hours (~48 hours)

STEP 3 — TEST FIRST, THEN RE-TIME

48 hours on 8 shared machines is a large commitment, so we ran a 15-minute test first. It found the problems on page 3. After fixing them the measured speed went from 1.39 to about 9.7 restarts per second per GPU — roughly 7x faster.

dataset	first estimate	after fixes	real time
MATH-500	138 GPU-h	~28 GPU-h	~5 hours
AIME-2025	34 GPU-h	~7 GPU-h	~1 hour
GPQA-Diamond	210 GPU-h	~43 GPU-h	~5 hours

The 8 GPUs do not finish together, because some questions are far longer than others. The job is done only when the slowest one is, so we quote ~5 hours rather than the ~3.5 hour average.

What we found and fixed

Problems that only appeared at the larger scale

1. ASKING FOR 100 RESTARTS AT ONCE DID NOT FIT IN GPU MEMORY

The GPU has to hold every in-progress restart in memory at the same time. Ten fitted comfortably; a hundred needed more than twice the space available, so the GPU spent its time shuffling data instead of generating — memory reserved, but 0% of its compute in use. Fixed by requesting the restarts in batches of 20.

This one mattered most: unfixed, all 8 GPUs would have run overnight and produced nothing. The trial run never showed it, because 10 restarts fitted easily.

2. THE FIRST STAGE KEPT THE GPUS ALMOST IDLE

The initial reasoning attempts were generated one question at a time, giving the GPU only 3 pieces of work at once — a small fraction of what it can handle. Doing all questions together made it ~5x faster; with 8 GPUs, this stage for MATH-500 went from an estimated 3.5 hours to about 5 minutes.

3. A SETTING WAS LEFT OVER FROM THE SMALL TRIAL

A limit on how much the GPU handles at once was still set for the 10-restart trial. Raising it to match the new workload gave a further 1.9x speed-up.

4. AIME'S LENGTH LIMIT WAS DISTORTING THE RESULTS, NOT JUST THE COST

More than half of AIME's reasoning attempts were still unfinished when they hit the length limit, and had to be discarded. Doubling the limit recovered most:

	old limit	new limit
usable attempts	41 (46%)	73 (81%)
accuracy	30.0%	42.2%

The accuracy rise is not the model improving. The old limit was cutting off exactly the hard problems that need long reasoning, so our earlier AIME score of 30% was too low — an artifact of our setup rather than a property of the model.

5. WHY GPQA-DIAMOND WAS DEFERRED

GPQA costs about 4x more per restart than MATH-500, because the model writes far more text on it. That is a difficulty effect: the model is trained mainly on maths and code, while GPQA is graduate biology, chemistry and physics, so it deliberates instead of settling. It was set aside when it looked like 210 GPU-hours; at ~43 it is now affordable and could be added back.